

ENGINEERING DRAWING

EDR 101

PLANE GEOMETRY

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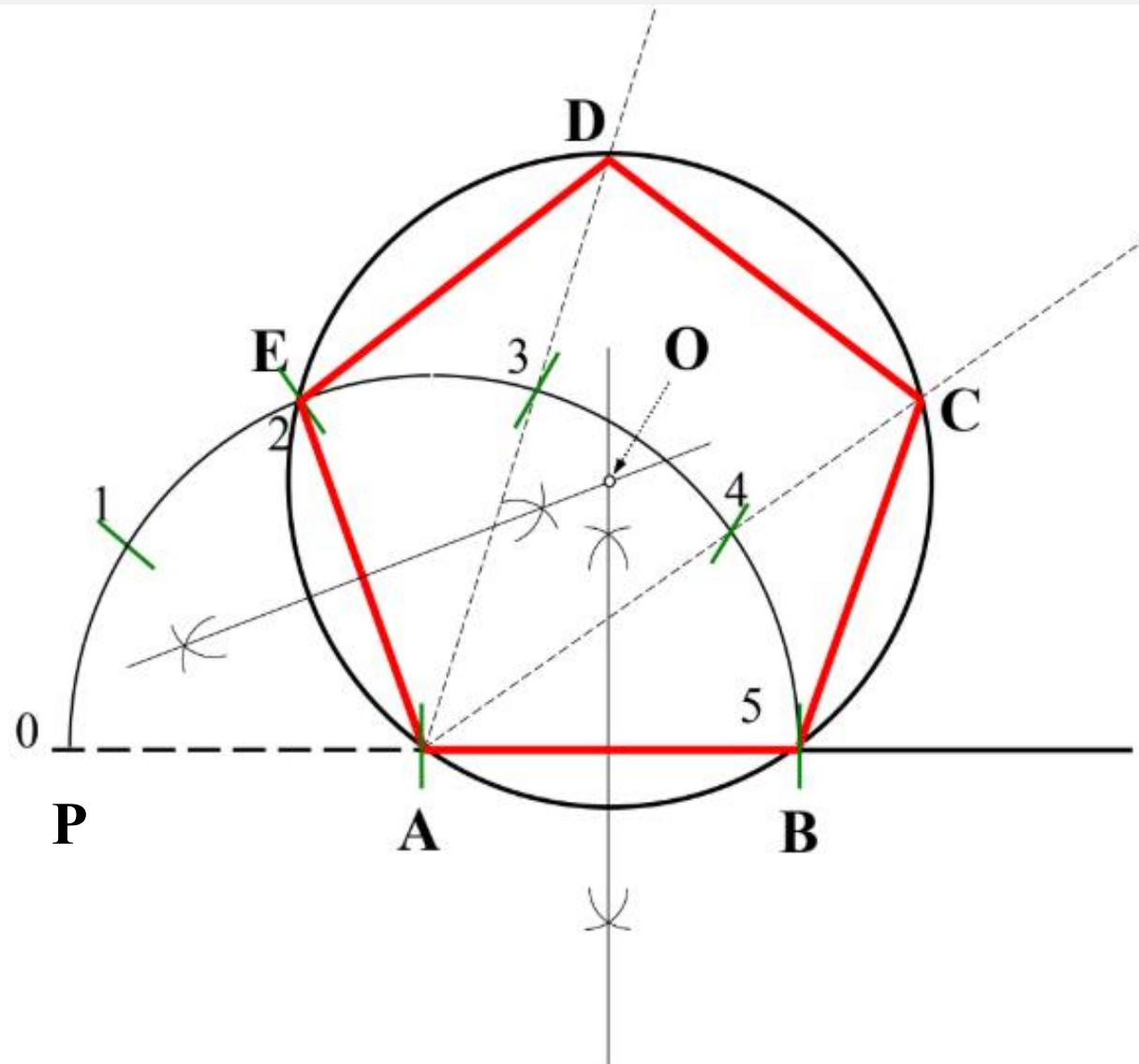
Department of Civil Engineering

Presidency University, Dhaka

COURSE CONTENT

Lecture	Topics
01	Introduction to Engineering Drawing
02	Lettering, Numbering, Dimensioning, Drawing Scales
03	Plane Geometry: Pentagon, Hexagon, Octagon, Ellipse, Parabola, Hyperbola
04	Solid Object: Polyhedra, Prism, Pyramid, Cone, Cylinder, Sphere, Frustum, Truncated
05	Projection: Orthographic, Isometric, Oblique
06	Drawing isometric projections of solids; Drawing isometric views of objects
MID TERM – I	
07	Drawing orthographic projections of solids and different objects
08	Introduction to AutoCAD: Interface, Basic command, Layout, Creating different types of drawing, Dimension & Annotation, Printing & Publishing
09	Course wrap-up and evaluation
FINAL EXAMINATION	

PENTAGON

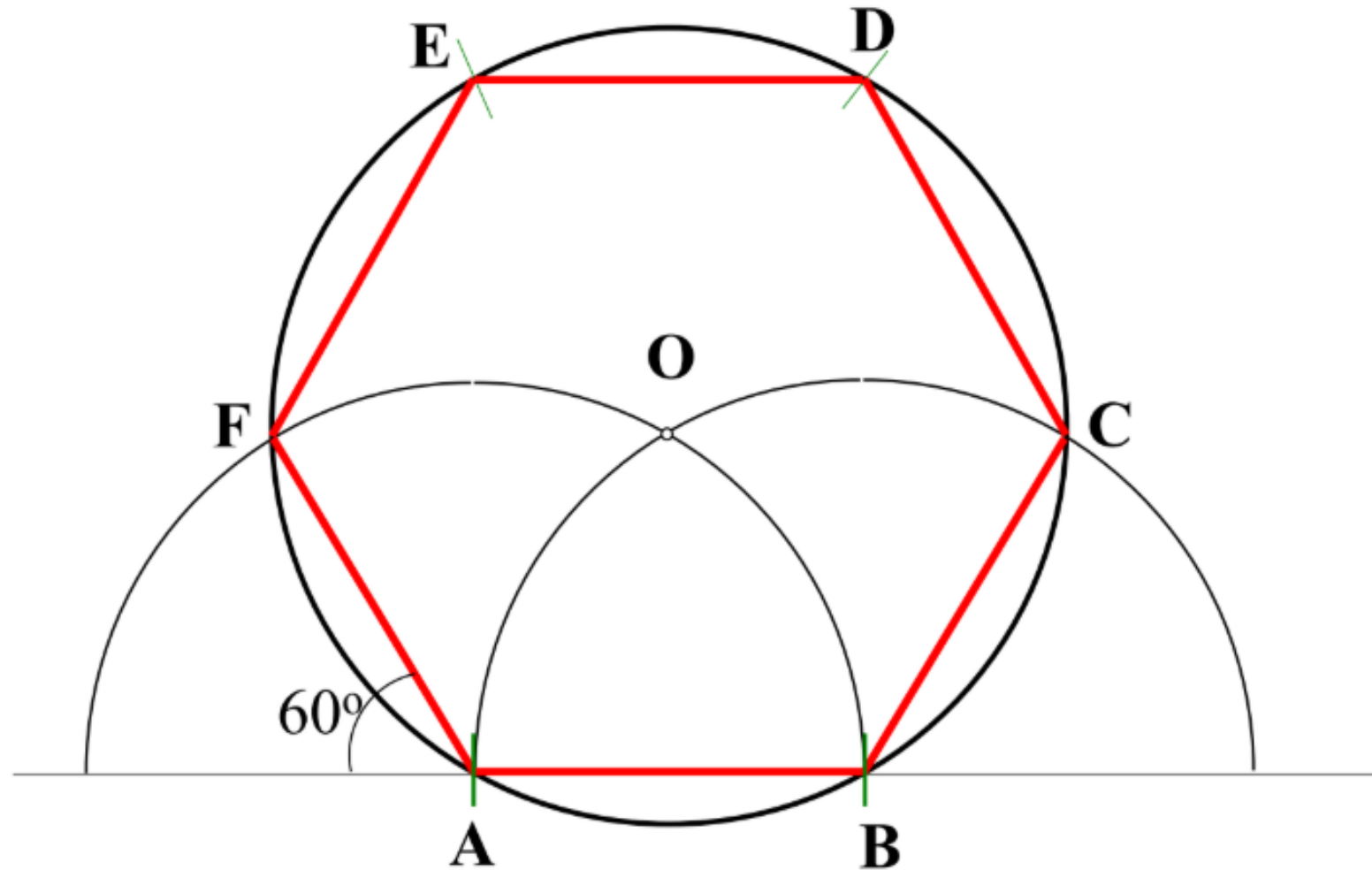


PENTAGON

To construct a regular pentagon given the length of the side –

1. Draw a line AB equal to the side and extend to P such that $AB = AP$
2. Draw a semicircle on PB and divide it into 5 equal parts by using a protractor and a compass ($180^\circ \div 5 = 36^\circ$)
3. Join A to the second division, which is point 2 on the semicircle (Mark it as E)
4. Irrespective of the number of sides of the polygon, A is always joined to the second division
5. Draw the perpendicular bisectors of AB and AE to intersect at O
6. Draw a circle with O as centre and OA as radius
7. With AB as chord, intersect the circle successively at C and D. Then join BC, CD, and DE

HEXAGON

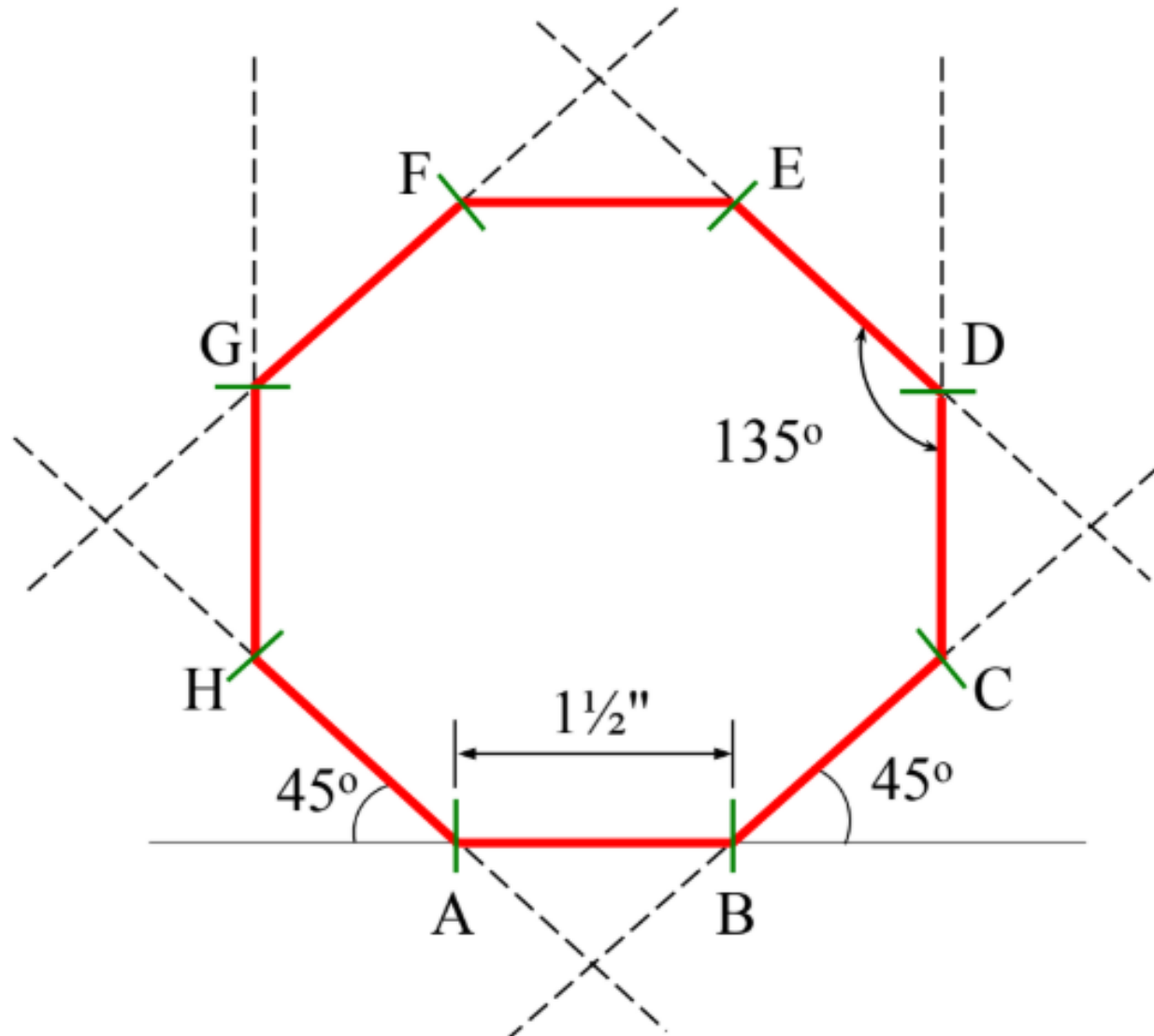


HEXAGON

To construct a regular hexagon given the length of the side –

1. Draw a line AB equal to the side, and draw two semicircles at points A and B on the extended AB line. The semicircles cross each other at point O
2. Draw a circle with O as centre and OA or OB as radius
3. The circle intersects the semicircles at F and C
4. With AB as chord, intersect the circle at E and D
5. Join BC, CD, DE, EF, and AF

OCTAGON

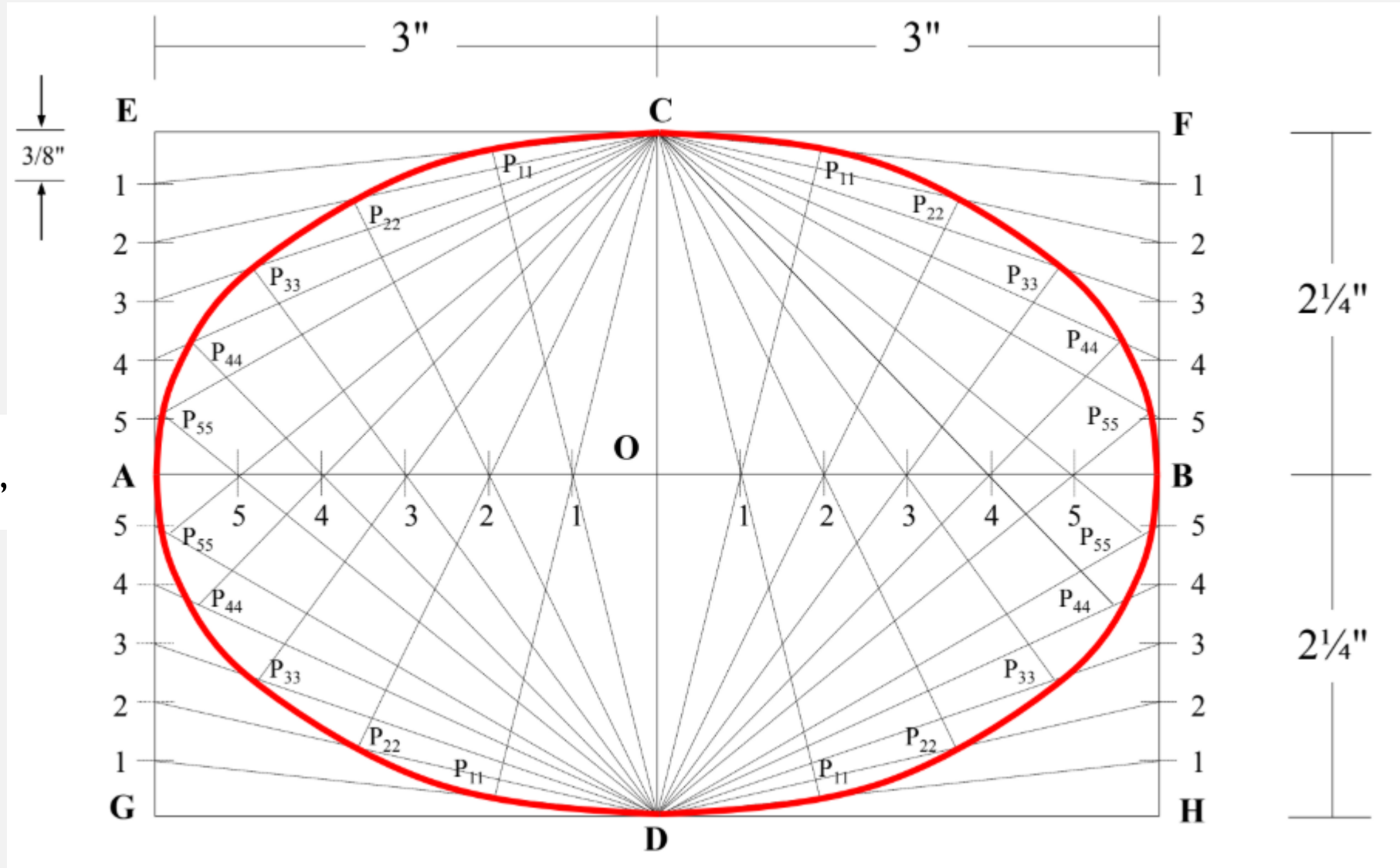


OCTAGON

To construct a regular octagon given the length of the side –

1. Draw a line AB equal to the side
2. Draw 45° exterior angle at points A and B. Then intersect at H and C, respectively, so that $AB=AH=BC$
3. Repeat No. 2 at points H and C, followed by points G and D
4. Join BC, CD, DE, EF, FG, and GH
5. Both 45° exterior angle and/or 135° interior angle can be drawn as seemed convenient in the above steps

ELLIPSE (PARALLELOGRAM METHOD)



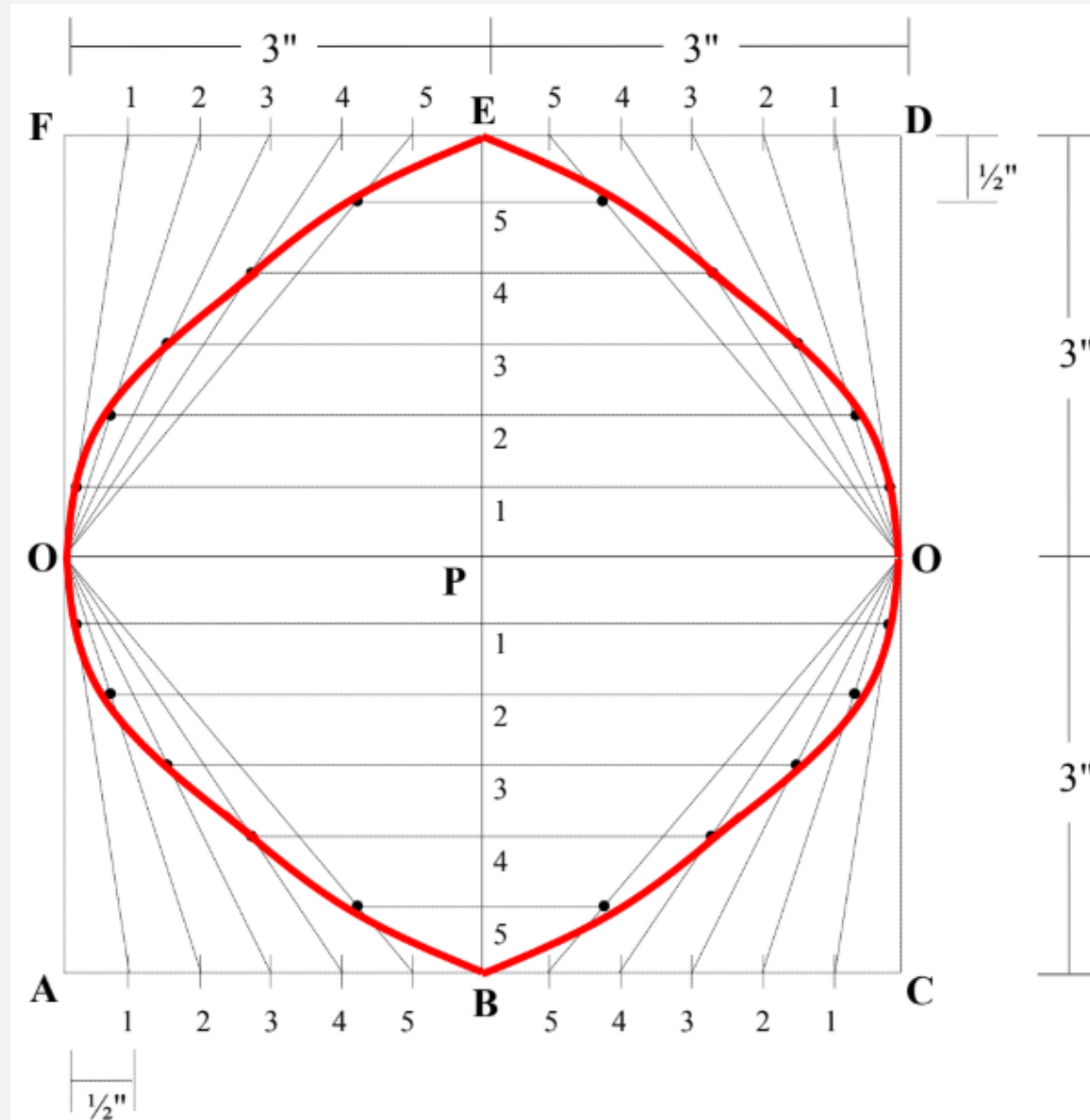
Major Axis, $AB = 6''$
Minor Axis, $CD = 4.5''$

ELLIPSE (PARALLELOGRAM METHOD)

To construct an ellipse given the length of the major and minor axes –

1. Draw the major and minor axes AB and CD, and locate the centre O
2. Draw the rectangle EFHG passing through A,D,B,C
3. Divide AO and AE into same number of equal parts, say 6
4. Join C with the points 1, 2, 3, 4, 5 on AE
5. Join D with the points 1, 2, 3, 4, 5 on AO and extend till they meet the lines C1, C2, C3, C4, C5 respectively at P11, P22, P33, P44, P55
6. Repeat steps 3 to 5 to obtain the points in the remaining three quadrants
7. Join the points by a smooth curve forming the required ellipse

PARABOLA (RECTANGULAR METHOD)

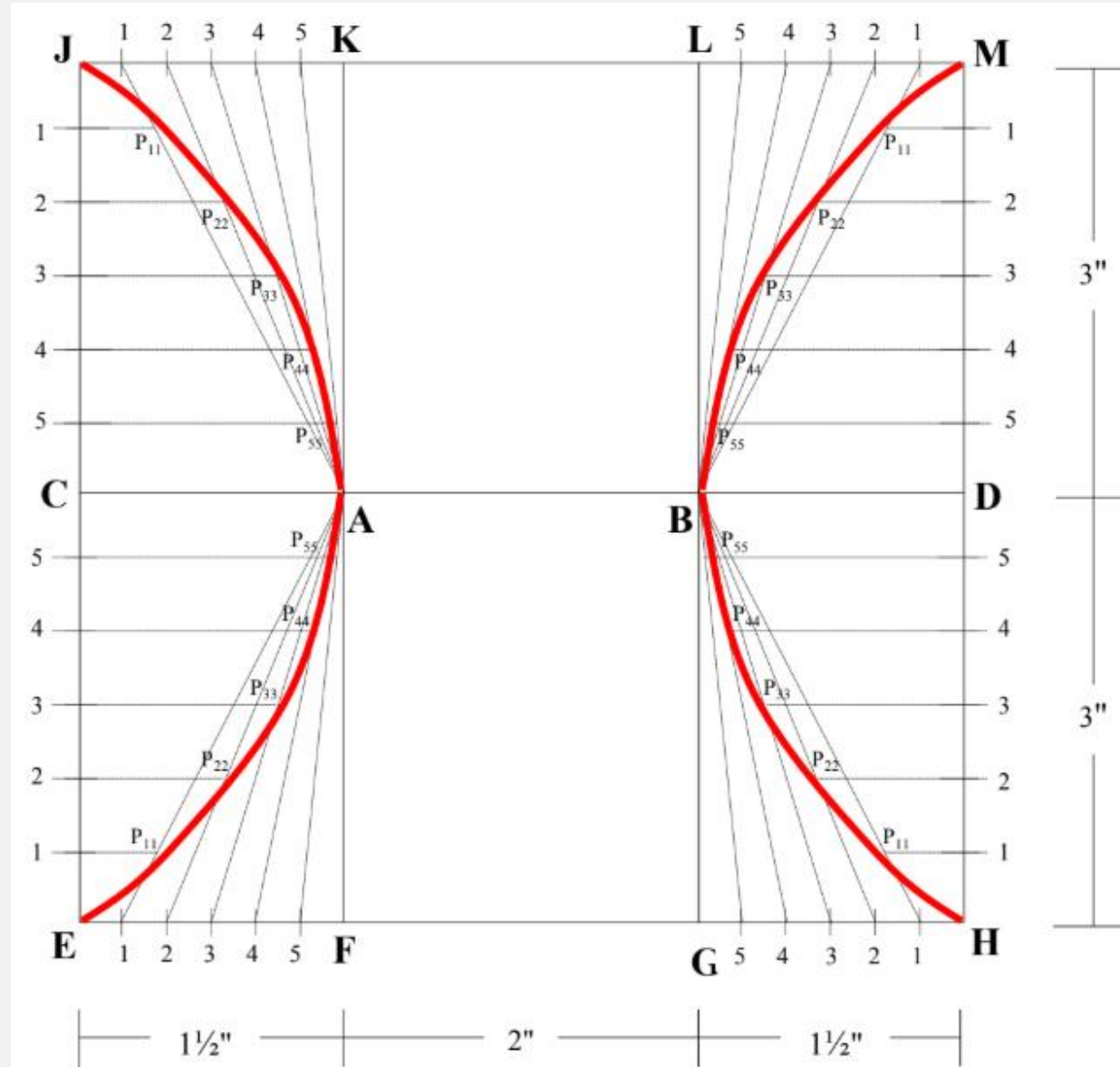


PARABOLA (RECTANGULAR METHOD)

To construct a parabola given the base and height (In the picture above, two parabolas are drawn on the base EB) –

1. Draw a line OP and EB perpendicular to each other. Here, EB is the base and OP is the height
2. Draw the rectangle ABEF
3. Divide FE, AB, PE, and PB into same number of equal parts, say 6
4. Join O with the points 1, 2, 3, 4, 5 on FE and AB
5. Draw lines from the points 1-5 on PE and PB parallel to PO and extend till they meet the lines O1-O5, respectively
6. Join the points by a smooth curve forming the required parabola

HYPERBOLA



THANK YOU